Sharp sphere-integral bounds for products of linear forms

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/ahl-197-on-the-existence-of-logarithmic-and-orbifold-jet-2024/pinned-original-source.tex (On the existence of logarithmic and orbifold jet differentials); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Statement at source lines 1594-1616, proof at lines 1618-1755 (single proof block, gate span-guard 4/4, batch gate 20/20 round1). Printed number 6.5 in Section 6, subsection Chern form inequalities (source line 1593). The counted span carries zero \cite and no bibliography need (the Landau-citing Remark sits outside the span at 1757+). The proof references item:Chern_curv_inequality2 (labeled enumerate item of prop:Chern_curv_inequalities at source line 1792, printed (b) via alphenumi) as a cross-span pointer for this lemma's own (b) item; the published article renders these as 6.5(b). Batch-3 rotation band second card (duty split 2026-10-15 19:0x).

Editorial note. The label lemm:Chern_form_inequalities names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemm environment in the article source.

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Editorial scope. the counted statement and its own proof environment only

Source reference. Original reference b: sources/ahl-197-on-the-existence-of-logarithmic-and-orbifold-jet-2024/pinned-original-source.tex, source lines 1792–1792. Navigation pointer invoked at line(s) 1694, 1700: the counted text cites that enumerate item of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the item marker the article itself prints for it (the item's own counter representation, e.g. the bare letter of an \alphenumi enumerate; in the source the citation typically composes \ref{<enclosing statement>}\eqref{<item>}, rendering as statement-number

plus item letter); the cited item is an allowed input and none of it is reproduced inside the span.

Lemma 6.5. *Let $\ell_j \in (\mathbb{C}^r)^*$, $1 \leqslant j \leqslant p$, be non-zero complex linear forms on $\mathbb{C}^r$, where $(\mathbb{C}^r)^* \simeq \mathbb{C}^r$ is equipped with its standard hermitian form, and let μ the rotation invariant probability measure on $\mathbb{S}^{2r-1} \subset \mathbb{C}^r$. Then*

$$I(\ell_1, \dots, \ell_p) = \int_{\mathbb{S}^{2r-1}} |\ell_1(u)|^2 \dots |\ell_p(u)|^2 \, d\mu(u)$$

satisfies the following inequalities:

$$I(\ell_1, \dots, \ell_p) \leq \tfrac{p!\,(r-1)!}{(p+r-1)!} \prod_{j=1}^{p} |\ell_j|^2, \tag{a}$$

and the equality occurs if and only if the ℓ_j are proportional;

$$I(\ell_1, \dots, \ell_p) \geq \tfrac{(r-1)!}{(p+r-1)!} \prod_{j=1}^{p} |\ell_j|^2, \tag{b}$$

and the equality occurs if and only if $p \leqslant r$ and the ℓ_j are pairwise orthogonal.

Proof. Denote by $d\lambda$ the Lebesgue measure on Euclidean space and by $d\sigma$ the area measure of the sphere. One can easily check that the projection

$$\mathbb{S}^{2r-1} \to \mathbb{B}^{2r-2}, \quad u = (u_1, \dots, u_r) \mapsto v = (u_1, \dots, u_{r-1}),$$

yields $d\sigma(u) = d\theta \wedge d\lambda(v)$ where $u_r = |u_r|\, e^{i\theta}$ [just check that the wedge products of both sides with $\frac{1}{2} d|u|^2$ are equal to $d\lambda(u)$, and use the fact that $d\theta = \frac{1}{2i}(du_r/u_r - d\bar{u}_r/\bar{u}_r)$], thus, in terms of polar coordinates $v = t\,u'$, $u' \in \mathbb{S}^{2r-1}$, we have $d\sigma(u) = d\theta \wedge t^{2r-3}\, dt \wedge d\sigma'(u')$, and going back to the invariant probability measures μ on $\mathbb{S}^{2r-1}$ and μ' on $\mathbb{S}^{2r-3}$, we get $|u_r|^2 = 1 - |v|^2 = 1 - t^2$ and an equality

$$d\mu(u) = \tfrac{2r-2}{2\pi}\, d\theta \wedge t^{2r-3}\, dt \wedge d\mu'(u'). \tag{6.1}$$

If $\ell_1, \dots, \ell_p$ are independent of u_r, (6.1) and the Fubini theorem imply by homogeneity

$$\int_{\mathbb{S}^{2r-1}} |\ell_1(u')|^2 \dots |\ell_p(u')|^2 \, d\mu(u) = \tfrac{r-1}{p+r-1} \int_{\mathbb{S}^{2r-3}} |\ell_1(u')|^2 \dots |\ell_p(u')|^2 \, d\mu'(u'), \tag{6.2}$$

$$\int_{\mathbb{S}^{2r-1}} |\ell_1(u')|^2 \dots |\ell_{p-1}(u')|^2 \, |u_r|^2 \, d\mu(u) = \tfrac{r-1}{(p+r-2)(p+r-1)} \int_{\mathbb{S}^{2r-3}} |\ell_1(u')|^2 \dots |\ell_{p-1}(u')|^2 \, d\mu'(u') \tag{6.2$'$}$$

(for instance, in case (6.2$'$), we have to integrate $t^{2p-2}(1-t^2) \times t^{2r-3}\, dt$). For $p \leqslant r$, the formulas

$$\int_{\mathbb{S}^{2r-1}} |u_1|^{2p} \, d\mu(u) = \tfrac{p!\,(r-1)!}{(p+r-1)!}, \quad \int_{\mathbb{S}^{2r-1}} |u_1|^2 \dots |u_p|^2 \, d\mu(u) = \tfrac{(r-1)!}{(p+r-1)!},$$

are then obtained by induction on r and p.

1. For any $\ell \in (\mathbb{C}^r)^*$, we can find orthonormal coordinates on $\mathbb{C}^r$ such that $\ell(u) = |\ell|\, u_1$ in the new coordinates. Hence

$$\int_{\mathbb{S}^{2r-1}} |\ell(u)|^{2p}\, d\mu(u) = m_{r,p}\, |\ell|^{2p} \quad \text{where } m_{r,p} = \int_{\mathbb{S}^{2r-1}} |u_1|^{2p}\, d\mu(u) = \tfrac{p!\,(r-1)!}{(p+r-1)!}.$$

It follows from Hölder's inequality that

$$I(\ell_1, \ldots, \ell_p) \leqslant \prod_{j=1}^{p} \left(\int_{\mathbb{S}^{2r-1}} |\ell_j|^{2p}\, d\mu(u) \right)^{1/p} = m_{r,p} \prod_{j=1}^{p} |\ell_j|^2,$$

and that the equality occurs if and only if all ℓ_j are proportional.

2. We prove the inequality

$$I(\ell_1, \ldots, \ell_p) \geqslant \tfrac{(r-1)!}{(p+r-1)!} \prod_{j=1}^{p} |\ell_j|^2$$

by induction on p, the result being clear for $p = 0$ or $p = 1$. If we choose an orthonormal basis $(e_1, \ldots, e_r) \in \mathbb{C}^r$ such that $\ell_j(e_r) \neq 0$ for all j and replace ℓ_j by $(\ell_j(e_r))^{-1}\ell_j$, we can assume $\ell_j(e_r) = 1$. We then write $u = u' + u_r e_r$ with $u' \in e_r^{\perp} \simeq \mathbb{C}^{r-1}$ and

$$\ell_j(u) = \ell'_j(u') + u_r, \quad 1 \leqslant j \leqslant p, \quad \ell'_j \coloneqq \ell_{j|e_r^{\perp}}.$$

Let $s_k(\ell'_\bullet(u'))$ be the elementary symmetric functions in $\ell'_j(u')$, $1 \leqslant j \leqslant p$, with $s_0 \coloneqq 1$. We have

$$I(\ell_1, \ldots, \ell_p) = \int_{\mathbb{S}^{2r-1}} \prod_{j=1}^{p} \left|\ell'_j(u') + u_r\right|^2 d\mu(u) = \int_{\mathbb{S}^{2r-1}} \left| \sum_{k=0}^{p} s_k(\ell'_\bullet(u'))\, u_r^{p-k} \right|^2 d\mu(u).$$

We make a change of variable $u_r \mapsto u_r\, e^{i\theta}$ and take the average over $\theta \in [0, 2\pi]$. Parseval's formula gives

$$I(\ell_1, \ldots, \ell_p) = \int_{\mathbb{S}^{2r-1}} \sum_{k=0}^{p} \left|s_k(\ell'_\bullet(u'))\right|^2 |u_r|^{2(p-k)} d\mu(u),$$

and since

$$(2r-2) \int_0^1 t^{2k} (1-t^2)^{p-k}\, t^{2r-3} dt = \tfrac{(r-1)\,(k+r-2)!\,(p-k)!}{(p+r-1)!},$$

formula (6.1) implies

$$I(\ell_1, \ldots, \ell_p) = \int_{\mathbb{S}^{2r-3}} \sum_{k=0}^{p} \tfrac{(r-1)\,(k+r-2)!\,(p-k)!}{(p+r-1)!} \left|s_k(\ell'_\bullet(u'))\right|^2 d\mu'(u').$$

As $|\ell_j|^2 = 1 + |\ell'_j|^2$, our inequality 6.5(b) is equivalent to

$$\int_{\mathbb{S}^{2r-3}} \sum_{k=0}^{p} \tfrac{(k+r-2)!\,(p-k)!}{(r-2)!} \left|s_k(\ell'_\bullet(u'))\right|^2 d\mu'(u') \geq \prod_{j=1}^{p} \left(1 + |\ell'_j|^2\right) \qquad (6.3)$$

for all linear forms $\ell'_j \in (\mathbb{C}^{r-1})^*$. We actually prove (6.3) by induction on p (observing that the inequality is a trivial equality for $p = 0, 1$). Assume that (6.3) (and hence 6.5(b)) is known for any $(p-1)$-tuple of linear forms $(\ell'_1, \ldots, \ell'_{p-1})$. As 6.5(b) is invariant under the action of $U(r)$, it is sufficient to consider the case when $\ell_p(u) = u_r$, i.e. $\ell'_p = 0$. The induction hypothesis tells us that

$$\int_{\mathbb{S}^{2r-3}} \sum_{k=0}^{p-1} \tfrac{(k+r-2)!\,(p-1-k)!}{(r-2)!} \left|s_k(\ell'_\bullet(u'))\right|^2 d\mu'(u') \geq \prod_{j=1}^{p-1}(1+|\ell'_j|^2).$$

However, when we add the factor ℓ_p, the elementary symmetric functions $s_k(\ell'_\bullet(u'))$ are left unchanged for $k \leqslant p-1$, while $s_p(\ell'_\bullet(u')) = 0$ and $1+|\ell'_p|^2 = 1$. Therefore (6.3) holds true for p, since $(p-k)! \geqslant (p-1-k)!$ for all $k = 0, 1, \ldots, p-1$. We have proved the inequality at order p whenever $\ell_p = \alpha_p\langle \bullet, e_r\rangle$ and $\ell_j(e_r) \neq 0$ for $j \leqslant p-1$. Since those $(\ell_1, \ldots, \ell_p)$ are dense in the space $((\mathbb{C}^r)^*)^p$ of p-tuples of linear forms, the proof of the lower bound is complete.

3. (b, equality case) We argue by induction on r. For $r = 1$, we have in fact $\ell_j(u) = \alpha_j u_1$, $\alpha_j \in \mathbb{C}^*$, and $I(\ell_1, \ldots, \ell_r) = \prod|\ell_j|^2$, thus the coefficient $\frac{1}{(p+r-1)!} = \frac{1}{p!}$ is reached if and only if $p \leqslant 1$. Now, assume $r \geqslant 2$ and the equality case solved for dimension $r-1$. By rescaling and reordering the ℓ_j, we can always assume that $\ell_j(e_r) \neq 0$ (and hence $\ell_j(e_r) = 1$) for $q+1 \leqslant j \leqslant p$, while $\ell_j(e_r) = 0$ for $1 \leqslant j \leqslant q$ (we can possibly have $q = 0$ here). Then we write $\ell_j(u) = \ell'_j(u')$ for $1 \leqslant j \leqslant q$ and $\ell_j(u) = \ell'_j(u') + u_r$ for $q+1 \leqslant j \leqslant p$. Therefore, if $s_k(\ell'(u'))$ denotes the kth elementary symmetric function in $(\ell'_j(u')_{q+1\leqslant j\leqslant p}$, we find

$$\begin{aligned}
I(\ell_1, \ldots, \ell_p) &= \int_{\mathbb{S}^{2r-1}} \prod_{j=1}^{q} \left|\ell'_j(u')\right|^2 \prod_{j=q+1}^{p} \left|\ell'_j(u') + u_r\right|^2 d\mu(u) \\
&= \int_{\mathbb{S}^{2r-1}} \prod_{j=1}^{q} \left|\ell'_j(u')\right|^2 \left|\sum_{k=0}^{p-q} s_k(\ell'(u'))\, u_r^{p-q-k}\right|^2 d\mu(u) \\
&= \int_{\mathbb{S}^{2r-1}} \prod_{j=1}^{q} \left|\ell'_j(u')\right|^2 \sum_{k=0}^{p-q} \left|s_k(\ell'(u'))\right|^2 |u_r|^{2(p-q-k)}\, d\mu(u) \\
&= \int_{\mathbb{S}^{2r-3}} \prod_{j=1}^{q} \left|\ell'_j(u')\right|^2 \sum_{k=0}^{p-q} \tfrac{(r-1)\,(k+r-2)!\,(p-q-k)!}{(p-q+r-1)!} \left|s_k(\ell'(u'))\right|^2 d\mu'(u') \\
&\geqslant \tfrac{(r-1)!}{(p+r-1)!} \prod_{j=1}^{q} |\ell'_j|^2 \prod_{j=q+1}^{p} \left(1+|\ell'_j|^2\right)
\end{aligned}$$

by what we have just proved. In an equivalent way, we get

$$\begin{aligned}
\int_{\mathbb{S}^{2r-3}} \prod_{j=1}^{q} \left|\ell'_j(u')\right|^2 \sum_{k=0}^{p-q} \tfrac{(k+r-2)!\,(p-q-k)!\,(p+r-1)!}{(r-2)!\,(p-q+r-1)!} \left|s_k(\ell'(u'))\right|^2 d\mu'(u') \\
\geqslant \prod_{j=1}^{q} |\ell'_j|^2 \prod_{j=q+1}^{p} \left(1+|\ell'_j|^2\right)
\end{aligned}$$

In general, we can rotate coordinates in such a way that $\ell_p(u) = u_r$ and $\ell'_p = 0$, and we see that the above inequality holds when p is replaced by $p-1$, as soon as $q \leqslant p-2$. Then the corresponding coefficients $k = 0$ for p, $p-1$ are

$$\tfrac{(p-q)!\,(p+r-1)!}{(p-q+r-1)!} > \tfrac{(p-1-q)!\,(p-1+r-1)!}{(p-1-q+r-1)!},$$

and since $s_0 = 1$, we infer that the inequality is strict. The only possibility for the equality case is $q = p-1$, but then

$$I(\ell_1, \ldots, \ell_p) = \int_{\mathbb{S}^{2r-1}} \prod_{j=1}^{p-1} \left|\ell'_j(u')\right|^2 |u_r|^2 \, d\mu(u) = \tfrac{r-1}{p+r-1} \int_{\mathbb{S}^{2r-3}} \prod_{j=1}^{p-1} \left|\ell'_j(u')\right|^2 \, d\mu'(u'),$$

and we see that we must have equality in the case $(r-1, p-1)$. By induction, we conclude that $p-1 \leqslant r-1$ and that the $\ell_j(u) = \ell'_j(u')$ are orthogonal for $j \leqslant p-1$, as desired. □